



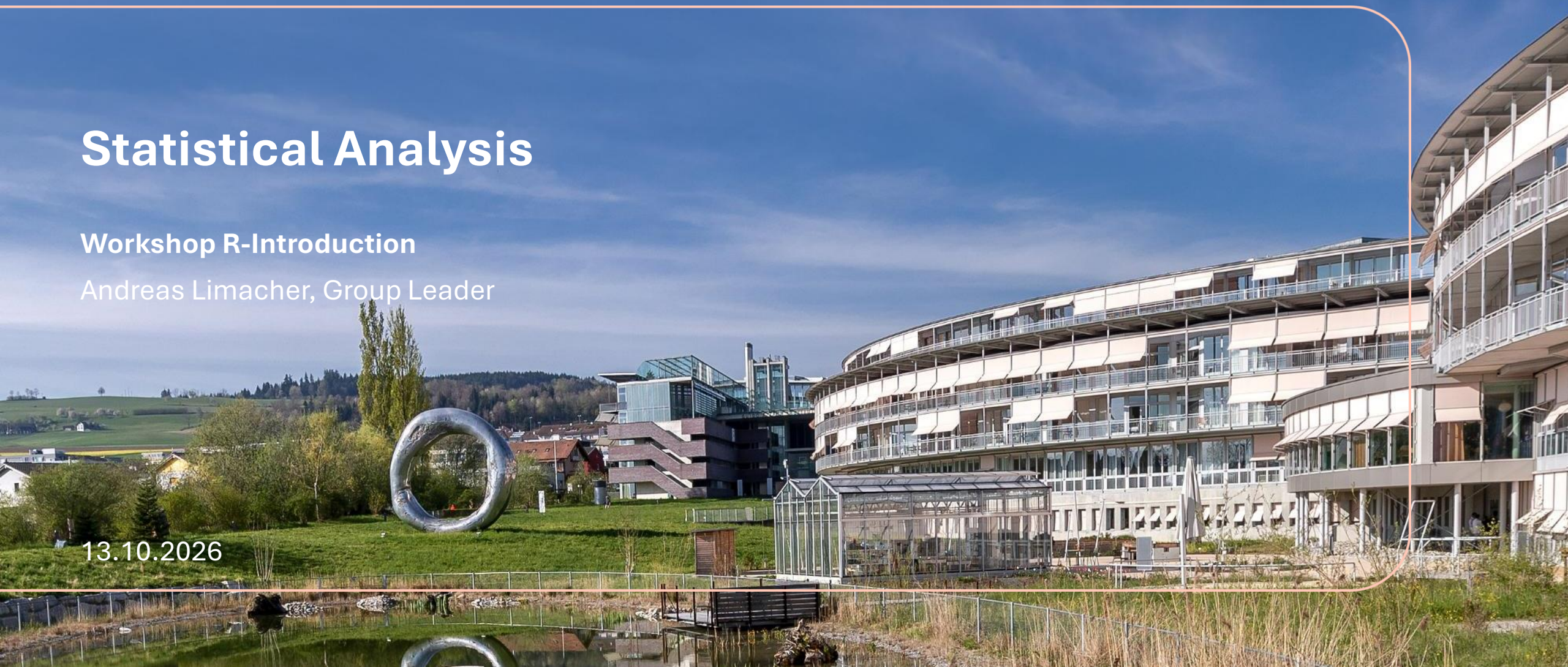
Swiss
Paraplegic
Research

Statistical Analysis

Workshop R-Introduction

Andreas Limacher, Group Leader

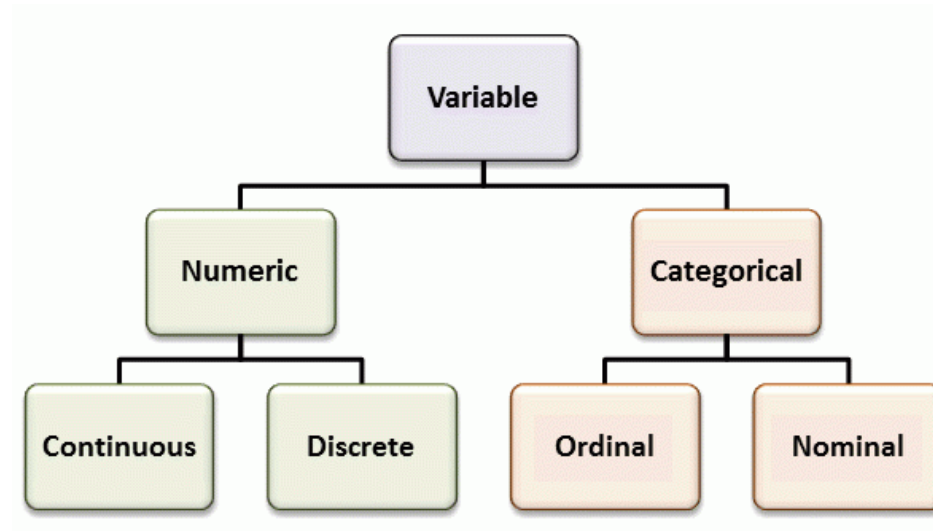
13.10.2026



Descriptive statistics

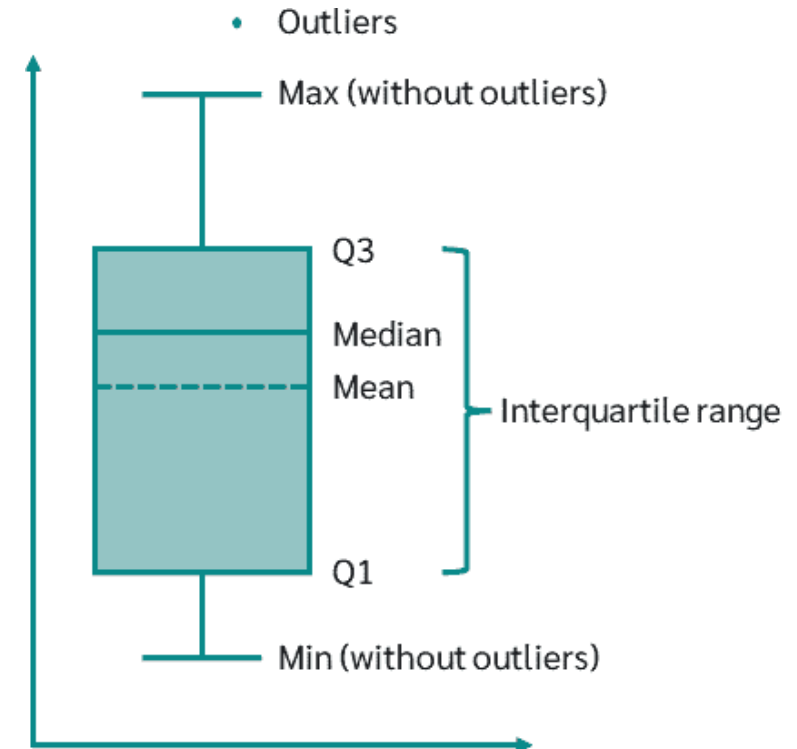


- What measures do you use to summarize data?
 - Categorical variables
 - Numeric variables



Descriptive statistics

- Continuous variables
 - Mean, standard deviation, confidence interval
 - Median, 1st quartile, 3rd quartile, interquartile range
- Categorical variables
 - Numbers, proportions



Descriptive statistics – Categorical

- Tabulation and cross-tabulation

```
> table(NHANES$Education, useNA = "always")
```

```
> table(NHANES$Education, NHANES$Gender, useNA = "always")
```

- Proportions

```
> prop.table(table(NHANES$Education))
```

```
> prop.table(table(NHANES$Education, NHANES$Gender), 2)
```

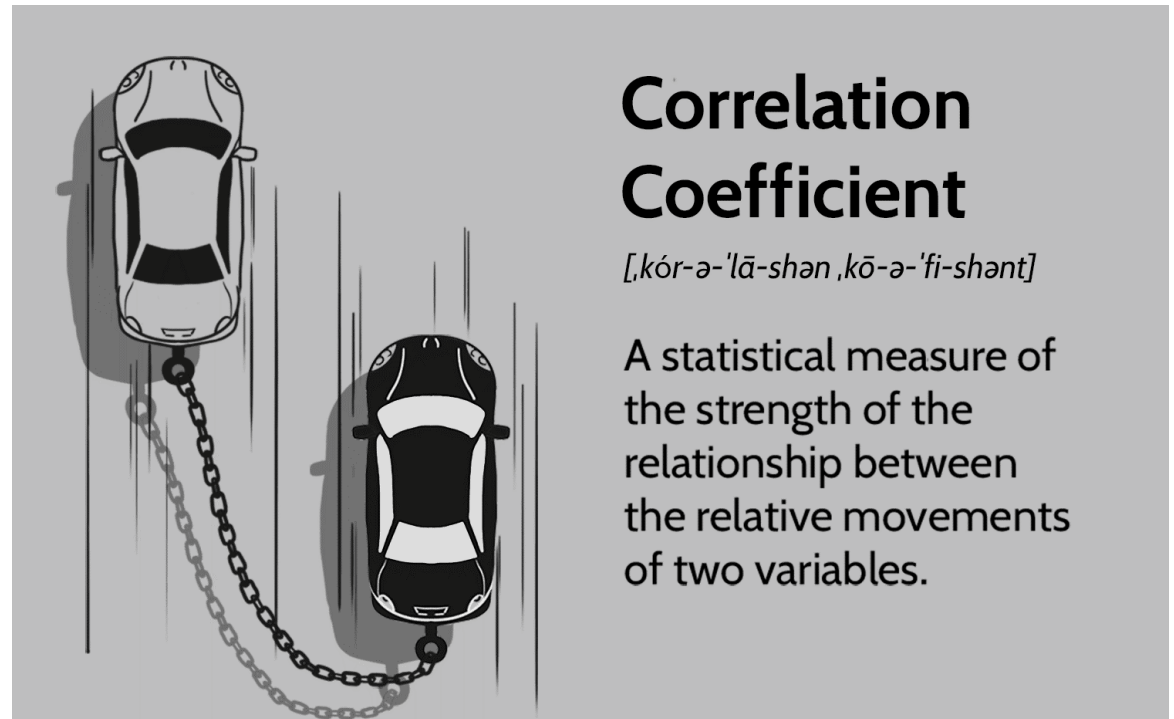
Descriptive statistics – Numeric

- Mean, sd, CI, median, and quartiles using tidyverse

```
> age_summary <- NHANES %>%  
  summarise(  
    mean_age = mean(Age, na.rm = TRUE),  
    sd_age = sd(Age, na.rm = TRUE),  
    n = sum(!is.na(Age)),  
    ci_lower = mean_age - qnorm(0.975) * sd_age / sqrt(n),  
    ci_upper = mean_age + qnorm(0.975) * sd_age / sqrt(n),  
    median_age = median(Age, na.rm = TRUE),  
    q1 = quantile(Age, 0.25, na.rm = TRUE),  
    q3 = quantile(Age, 0.75, na.rm = TRUE) )
```

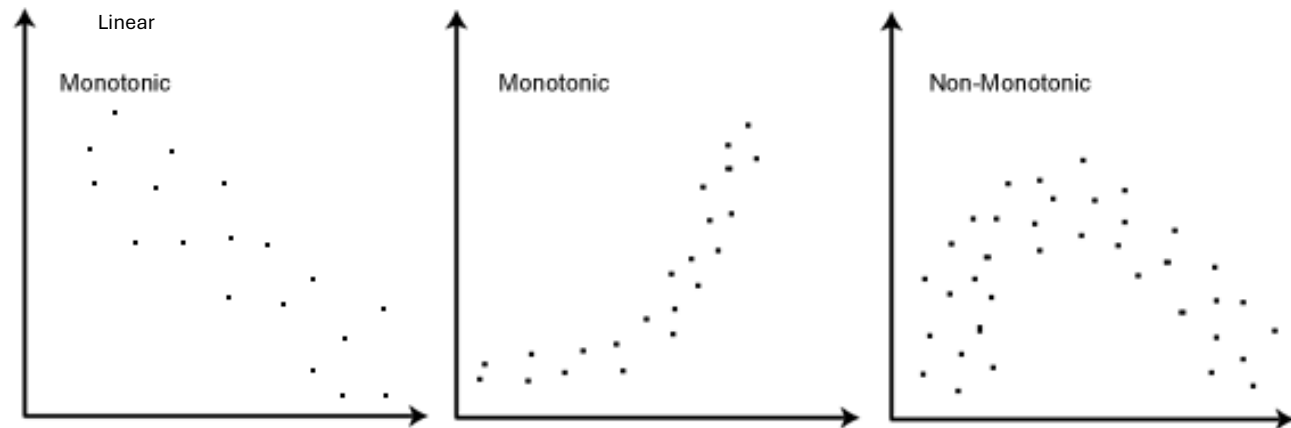
Correlation – Assumptions

- Which correlation coefficients can be used for numeric data?
- What are the assumptions?



Correlation – Assumptions

- Pearson correlation
 - Normal distribution (approximately)
 - Linear relationship between two variables
 - Observations are independent of each other
- Spearman correlation
 - Monotonic relationship between two variables
 - Observations are independent of each other



Correlation – R code

- Pearson correlation

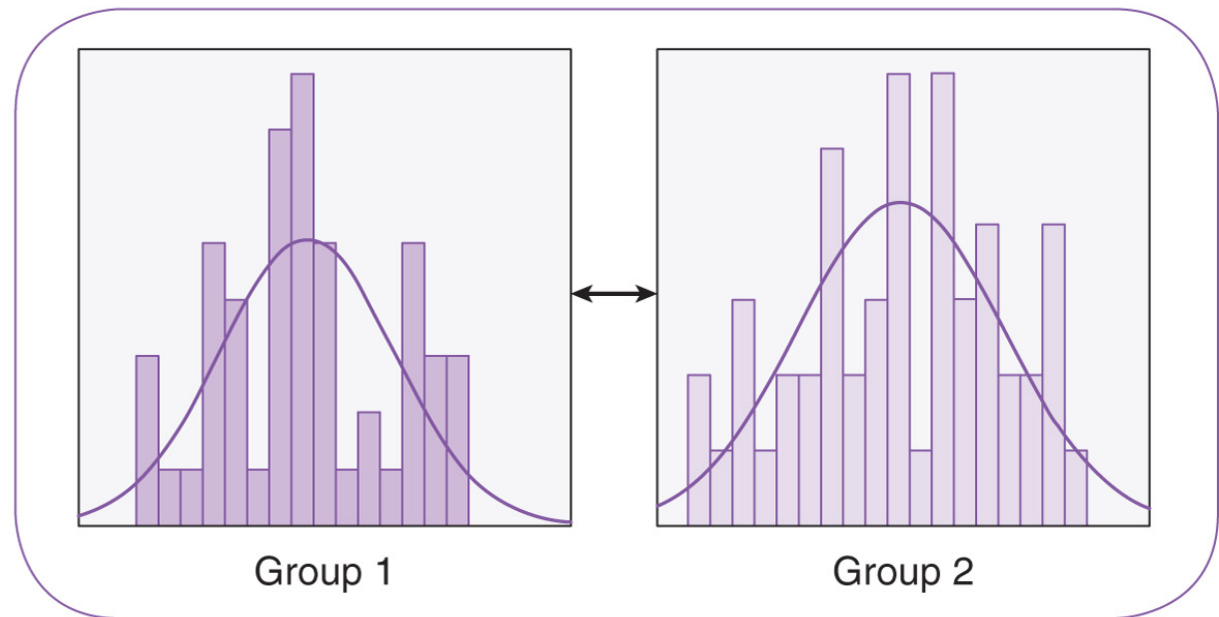
```
> NHANES %>%  
  select(Age, BMI) %>%  
  drop_na() %>%  
  summarise(correlation = cor(Age, BMI, method = "pearson"))
```

- Spearman correlation

```
> NHANES %>%  
  select(Age, BMI) %>%  
  drop_na() %>%  
  summarise(correlation = cor(Age, BMI, method = "spearman"))
```

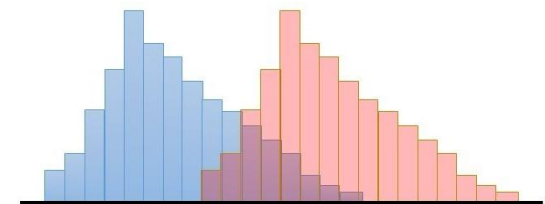
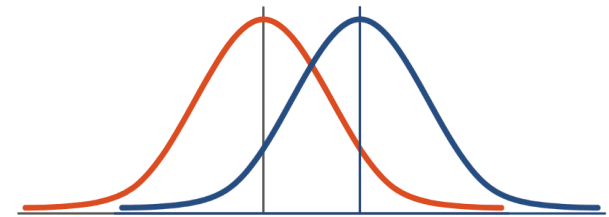
Common statistical tests

- Which tests do you know?
- When are they used?



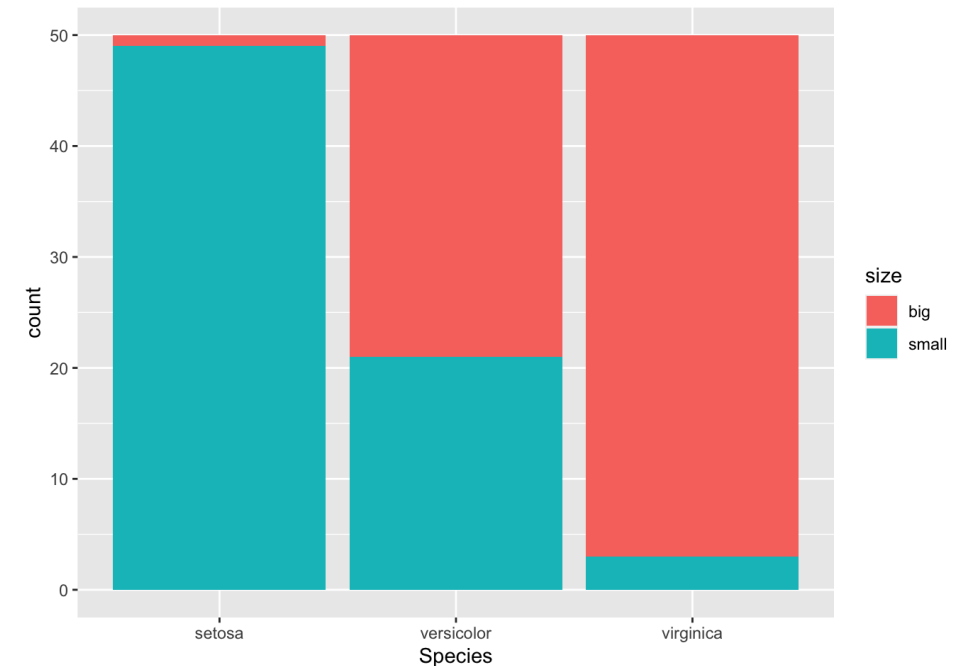
Common statistical tests - Numeric

- Normal assumption
 - T-test (one-group, 2 independent groups)
 - Paired t-test (2 paired groups)
 - ANOVA (multiple independent groups)
 - Repeated measure ANOVA (multiple paired groups)
- Non-parametric (no normal assumption)
 - Wilcoxon rank sum test / Mann-Whitney-U test (2 independent groups)
 - Wilcoxon signed rank test (2 paired groups)
 - Kruskal-Wallis test (multiple independent groups)
 - Friedman test (multiple paired groups)



Common statistical tests - Categorical

- Categorical data
 - Chi-square test (independent groups)
 - Fisher's exact test (independent groups, sparse data)
 - McNemar's test (paired data)



Common statistical tests – Numeric normal

- One-sample t-test (based on average US population age)

```
> t.test(NHANES$Age, mu = 38.5)
```

- Two-sample t-test

```
> t.test(Age ~ Gender, data = NHANES)
```

- Paired t-test

```
> t.test(NHANES$BPSys1, NHANES$BPSys2, paired = TRUE)
```

- ANOVA

```
> summary(aov(Age ~ Education, data = NHANES))
```



Common statistical tests – Numeric non-parametric

- Wilcoxon ranksum test

```
> wilcox.test(Age ~ Gender, data = NHANES)
```

- Wilcoxon signed rank test

```
> wilcox.test(NHANES$BPSys1, NHANES$BPSys2, paired = TRUE)
```

- Kruskal-Wallis test

```
> kruskal.test(Age ~ Education, data = NHANES)
```

Common statistical tests - Categorical

- Chi-square test

```
> chisq.test(table(NHANES$Gender, NHANES$Education))
```

- Fisher's exact test

```
> fisher.test(table(NHANES$Gender[1:50], NHANES$Education[1:50]))
```

- McNemar test for high blood pressure

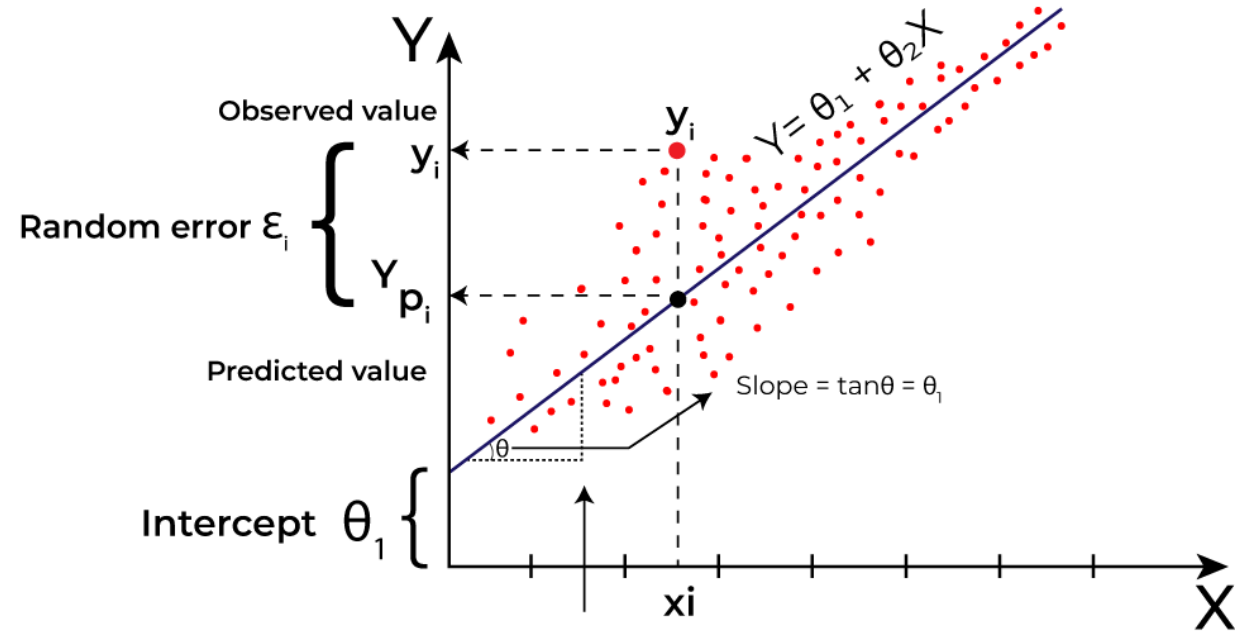
```
> bp_high1 <- NHANES$BPSys1 > 140
```

```
> bp_high2 <- NHANES$BPSys2 > 140
```

```
> mcnemar.test(bp_high1, bp_high2)
```

Linear regression

- Is age associated with BMI? Is there an influence of age on BMI?



Linear regression

```
> model <- lm(BMI ~ Age, data = NHANES)
```

```
> summary(model)
```

Residuals:

Min	1Q	Median	3Q	Max
-16.794	-4.803	-1.236	3.466	55.697

How do you interpret the model output?

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	21.451688	0.137193	156.36	<2e-16 ***
Age	0.138033	0.003148	43.84	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.735 on 9632 degrees of freedom

Multiple R-squared: 0.1664, Adjusted R-squared: 0.1663

F-statistic: 1922 on 1 and 9632 DF, p-value: < 2.2e-16

```
> confint(model, level = 0.95)
```



Linear regression

```
> model <- lm(BMI ~ Age, data = NHANES)
```

```
> summary(model)
```

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Difference between observed and predicted values

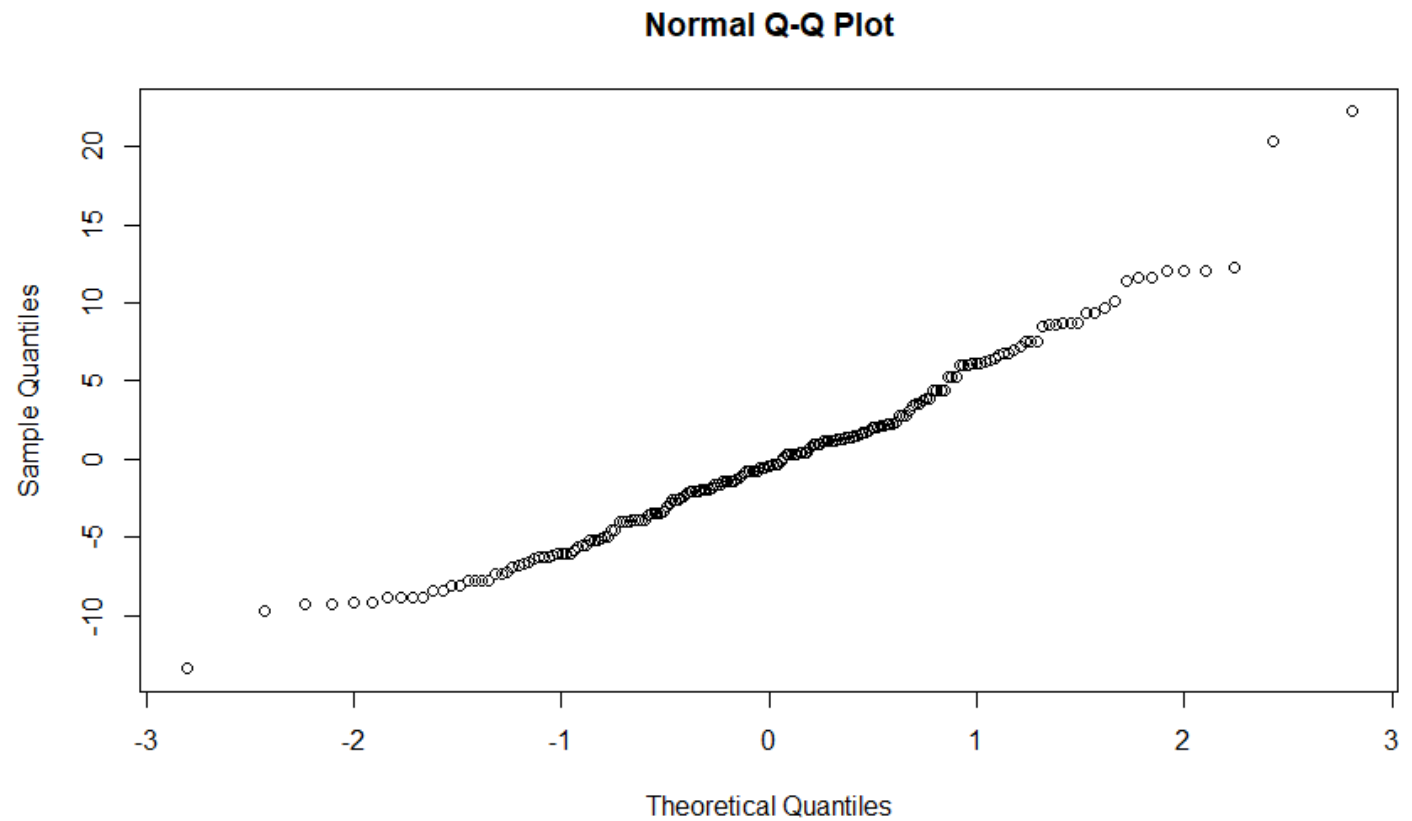
Change in dependent variable

BMI increases by 0.14 if age increases by 1 year

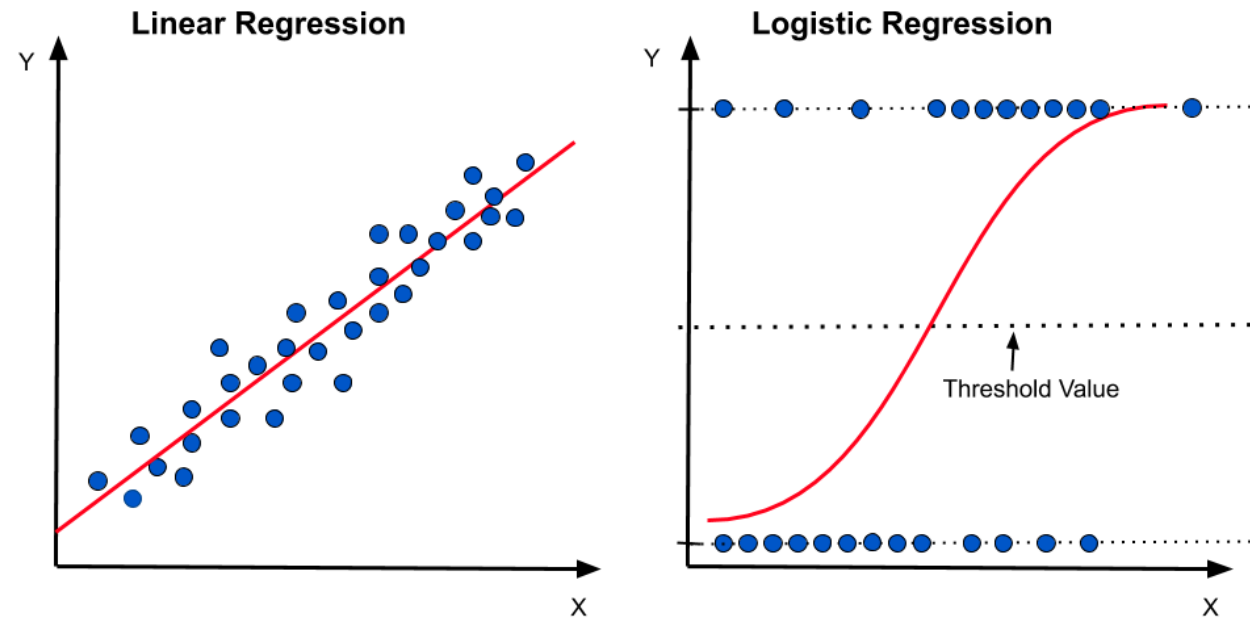
Variance explained by the model (%)
(adjusted for number of predictors)

Linear regression – Normal Q-Q Plot

```
> qqnorm(residuals(model))
```



Is age associated with obesity?



Logistic model

```
> model <- glm(HighBMI ~ Age_centered, data = NHANES, family = binomial)
> summary(model)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.988679	0.023903	-41.36	<2e-16 ***
Age_centered	0.023872	0.001084	22.01	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Null deviance: 11552 on 9633 degrees of freedom

Residual deviance: 11040 on 9632 degrees of freedom

AIC: 11044

How do you interpret the model output?

```
> exp(cbind(OR = coef(model), confint(model)))
```

	OR	2.5 %	97.5 %
(Intercept)	0.372068	0.3549663	0.3898363
Age_centered	1.024159	1.0219920	1.0263459

Logistic model

```
> model <- glm(HighBMI ~ Age_centered, data = NHANES, family = binomial)
> summary(model)
```

Coefficients:

Change on logit-scale

```
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -0.988679   0.023903  -41.36   <2e-16 ***
Age_centered  0.023872   0.001084   22.01   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Null deviance: 11552 on 9633 degrees of freedom

Residual deviance: 11040 on 9632 degrees of freedom

Unexplained variation (lower values indicate a better model fit)

AIC: 11044

Model fit adjusted for number of predictors

```
> exp(cbind(OR = coef(model), confint(model)))
```

OR 2.5 % 97.5 %

Relative change expressed as odds ratio

```
(Intercept)  0.372068 0.3549663 0.3898363
Age_centered 1.024159 1.0219920 1.0263459
```

Exercise

